NLP AND LANGUAGE MODELS

EdinburghAI Workshop 4



NATURAL LANGUAGE PROCESSING

- What are some AI tools that deal with language?

NATURAL LANGUAGE PROCESSING

- What are some AI tools that deal with language?
  - Chatbots like Claude and ChatGPT.
  - Machine Translation- DeepL, Google Translate etc.
  - Text summarizers
  - Sentiment Analysis
  - Text Generation and Writing Assistants- Grammarly.
  - Speech & Voice Tools- Alexa, Siri, Text-to-Speech.

HOW DOES A LANGUAGE MODEL WORK?



TOKENIZERS- MODELS NEVER SEE “WORDS”

- Breaks text into smaller units (subwords, characters)
- Handles unknown words by breaking them into known parts
- Each token maps to a unique number in the vocabulary

Text Input

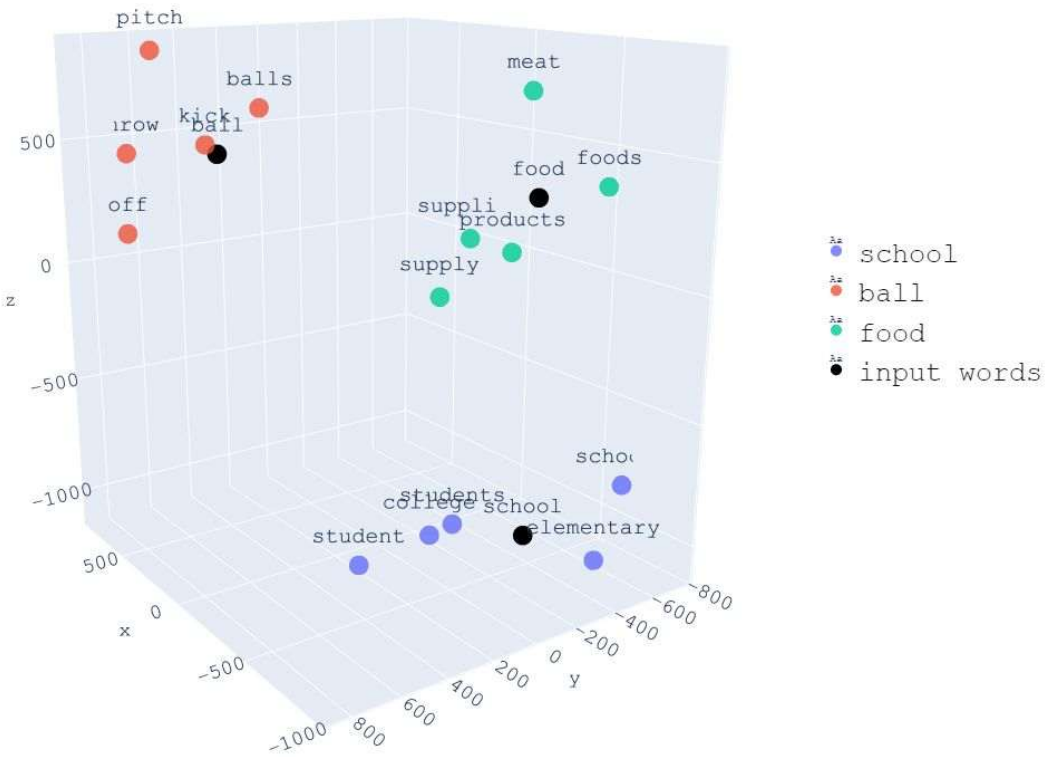
```
<|im_start|>What is the capital of France?<|im_sep|>
```

Tokens

```
200264, 4827, 382, 290, 9029, 328, 10128, 30, 200266
```

WORD EMBEDDINGS

- Each token gets mapped to a high-dimensional vector that represents its meaning.
- Words with similar meaning are close to one another.
- Relationships between words are represented geometrically.



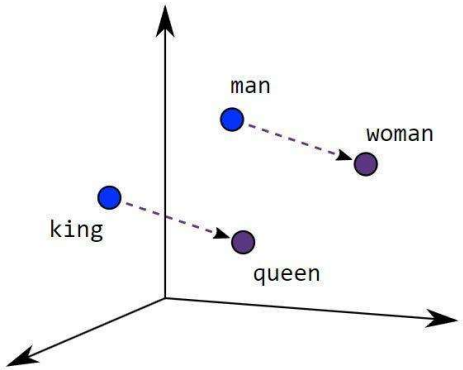
VECTOR MATH

- Consider the vectors $\vec{F}$ and $\vec{P}$ which represent the words “France” and “Paris”.
- If we subtract $\vec{F}$ from $\vec{P}$, we should get a vector that roughly corresponds to the concept “the capital city of”.

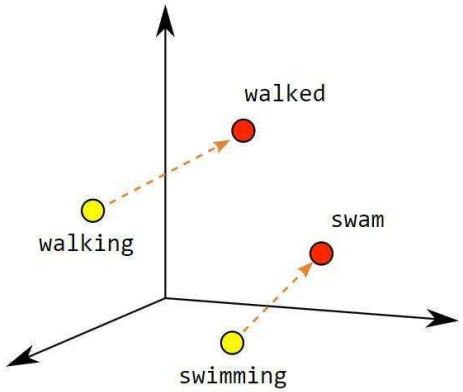
$$\vec{B} \approx \vec{G} + (\vec{P} - \vec{F})$$

- Where $\vec{B}$ and $\vec{G}$ represent “Berlin” and “Germany” respectively.
- In short: Berlin is to Germany what Paris is to France

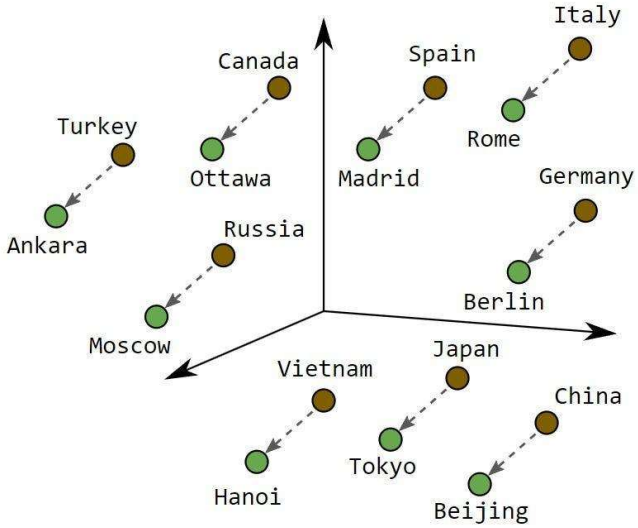
VECTOR RELATIONSHIPS



Male-Female



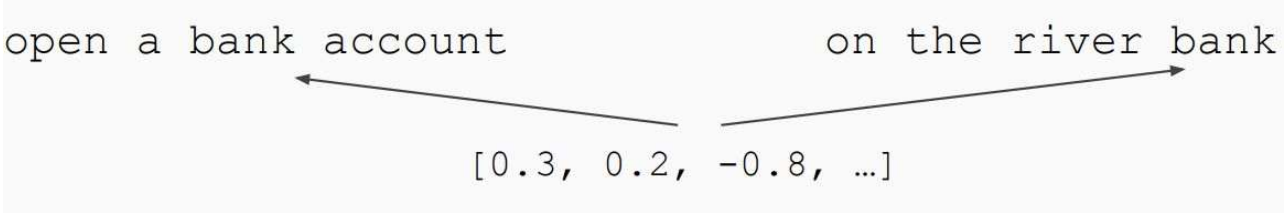
Verb Tense



Country-Capital

CONTEXTUAL EMBEDDINGS

- Word Embeddings are super useful, but they don't take into account the context a word is in.



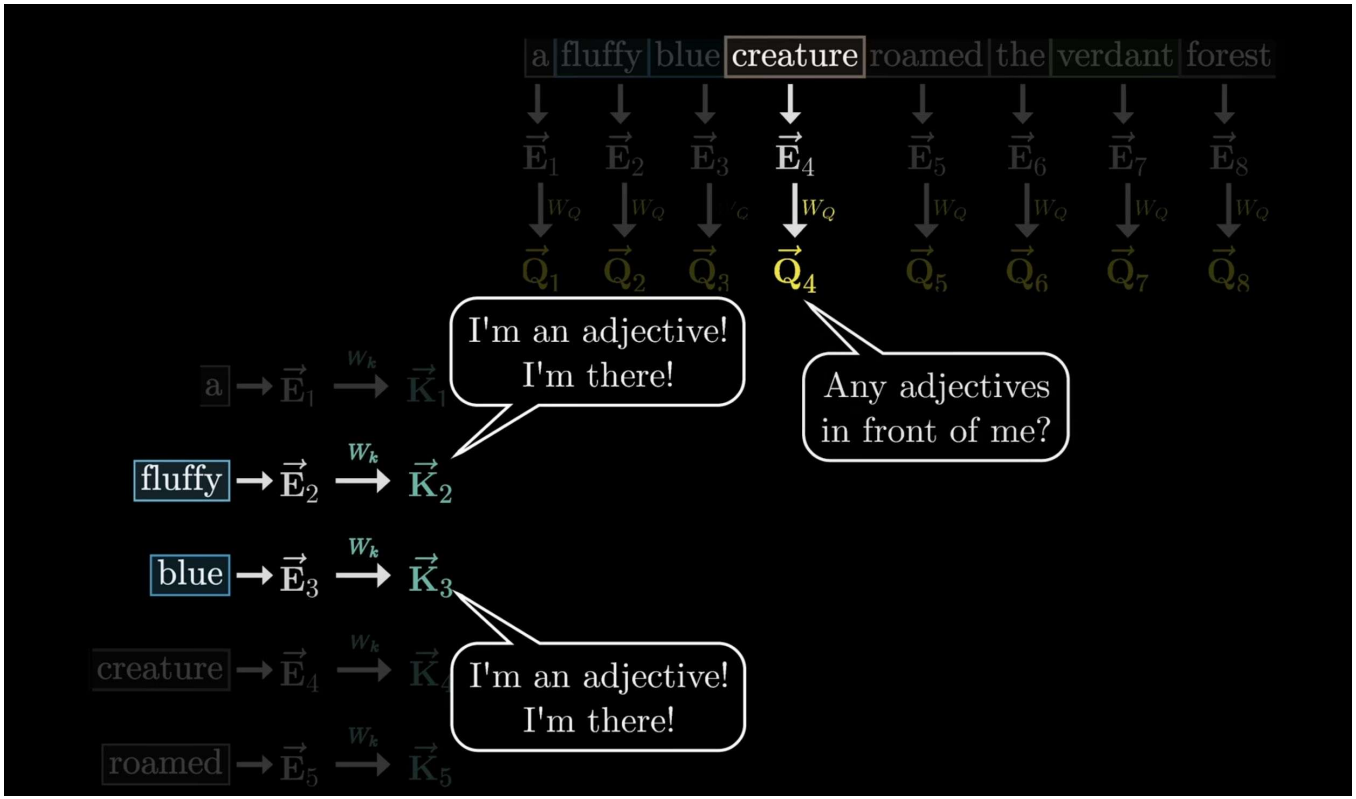
- So, we train models that take these word embeddings and contextualize them.



A SUPER-HIGH LEVEL OVERVIEW OF ATTENTION

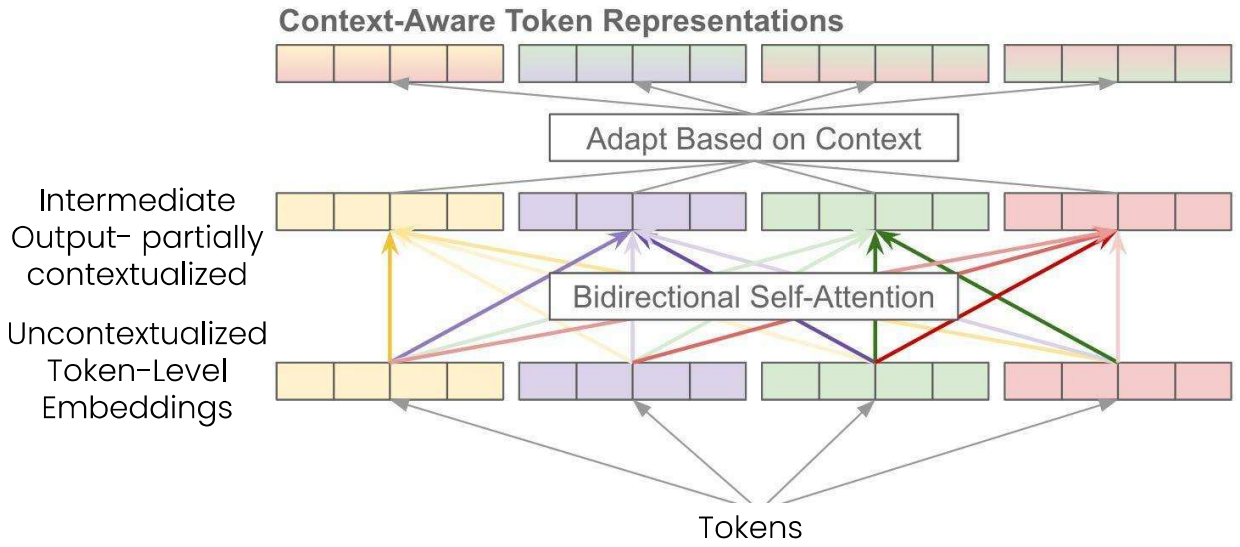
- Imagine a conversation:
- Each person asks a question- that's their **Query**.
- Everyone else responds with two things:
  - **Key:** "Does what I know answer that question?"
  - **Value:** "Here's the information I can share."
- The question-asker compares their **Query** to everyone's **Keys**.
- The better the match, the more they listen to that person's **Value**.
- After hearing everyone, they update their own understanding based on the weighted responses.

APPLYING THIS TO TOKENS



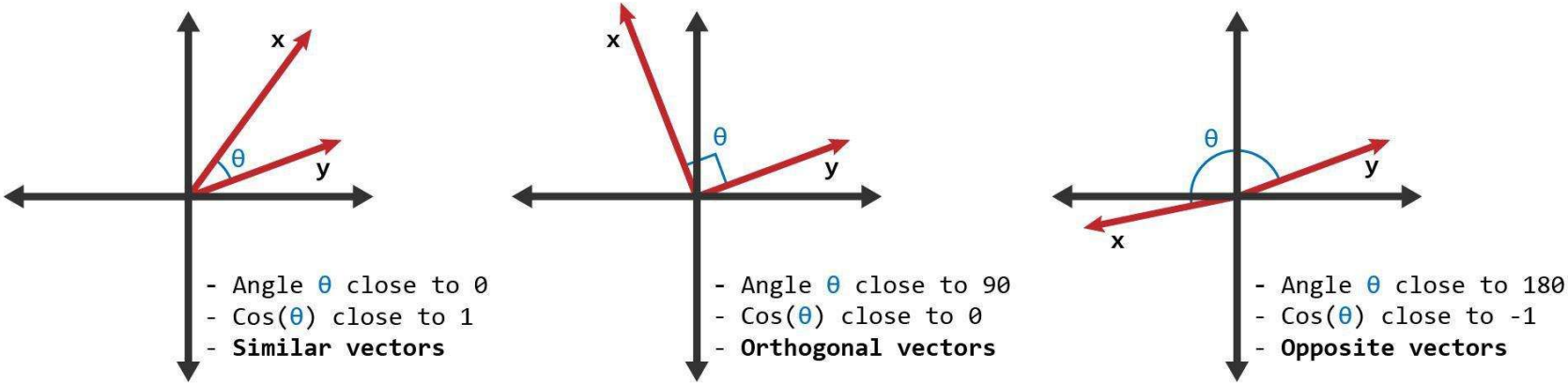
BERT MODELS

- BERT maps uncontextualized word embeddings → contextualized embeddings
- Achieves this using attention
- Pre-trained on a huge corpus of text.



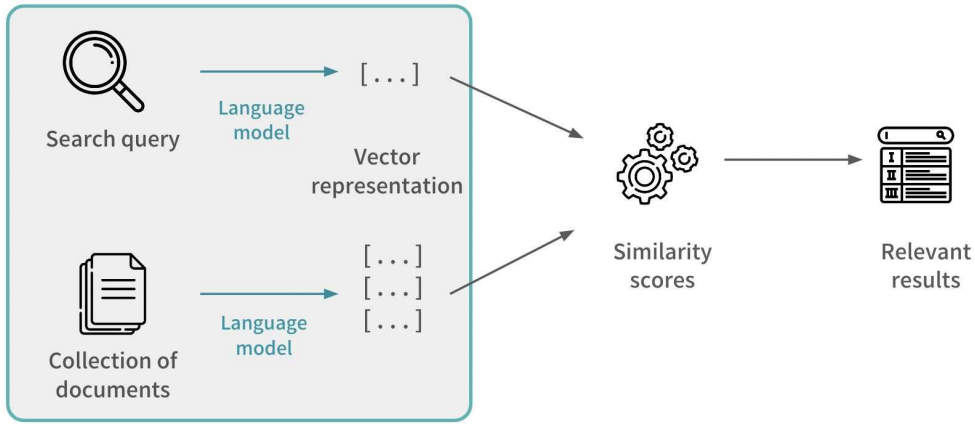
MEASURING SEMANTIC SIMILARITY

- $$\cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2} \sqrt{\sum_{i=1}^{n} B_i^2}}$$

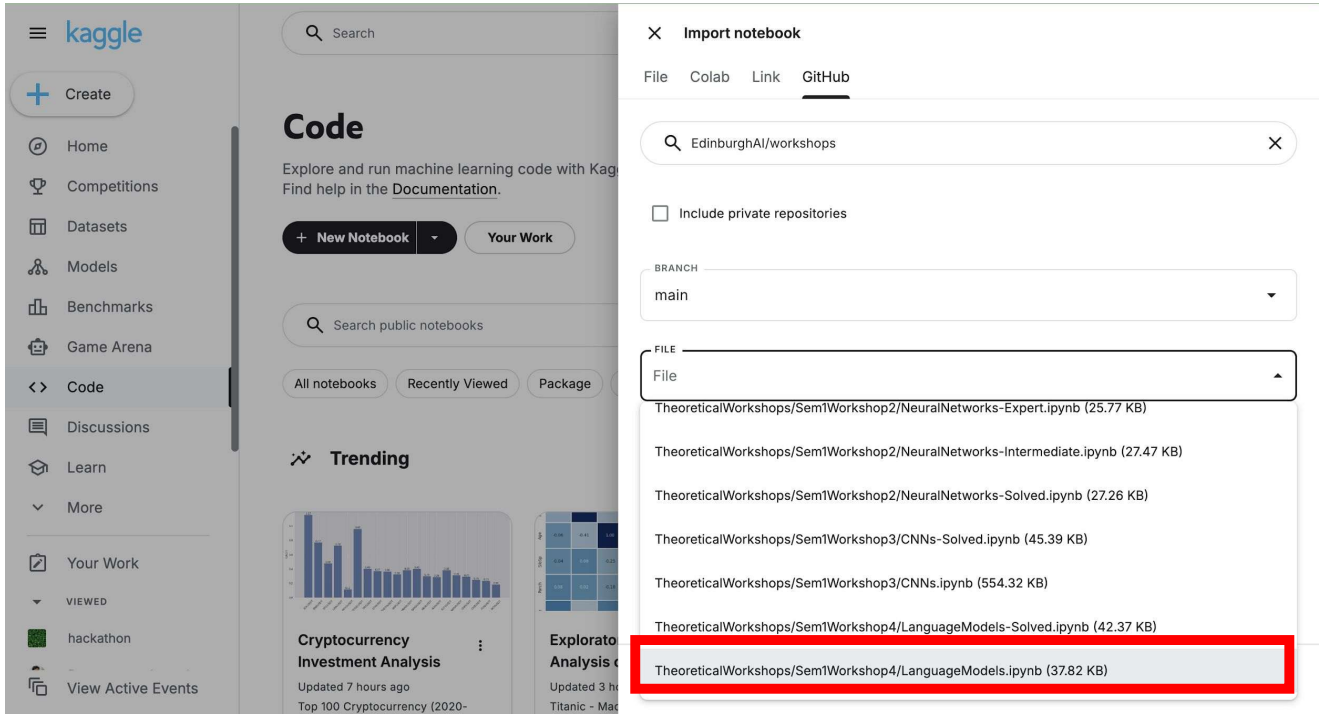


SEMANTIC SEARCH

1. Embed Documents and store the embeddings in a **vector database**.
2. Embed the search query.
3. Search the database for relevant items by computing similarity scores.
4. Produce results.



OPEN THE KAGGLE NOTEBOOK AND COMPLETE THE EMBEDDINGS AND SEMANTIC SEARCH TASKS!



FINE-TUNING A MODEL

- In the previous semantic search task, we made use of a pre-trained model.
- What if we wanted to use an existing model but for a specific task like sentiment analysis or text classification?
- Fine-tuning lets us adapt the model to our domain with just a small dataset.

PRE-TRAINING VS FINE-TUNING

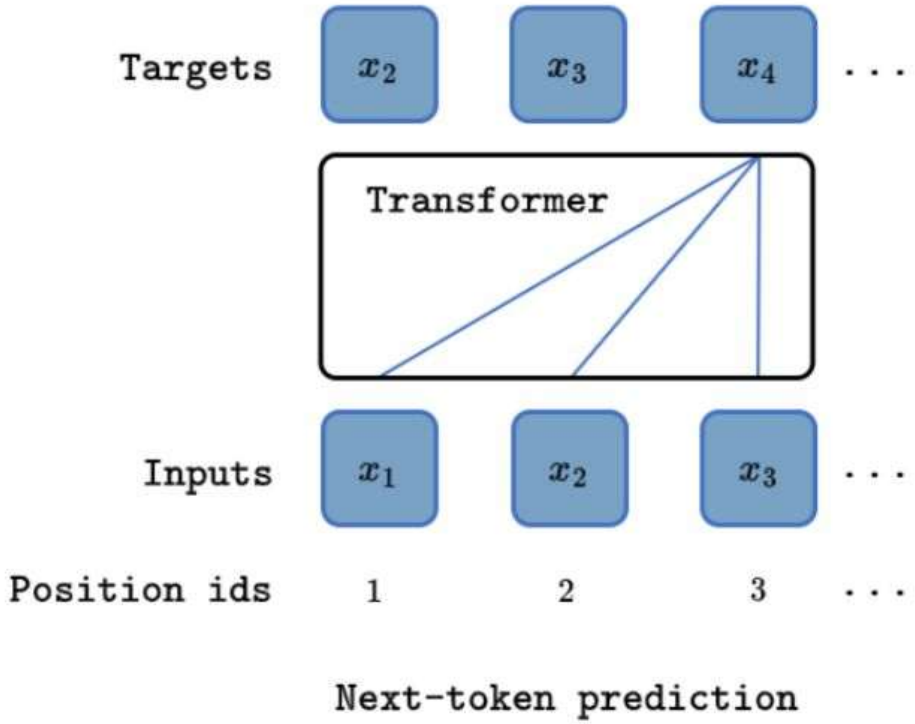
Pre-training is expensive and slow

- Train on billions of words
- Learn general language patterns
- Takes weeks/months on powerful GPUs
- Done once by companies like OpenAI, Anthropic, Google

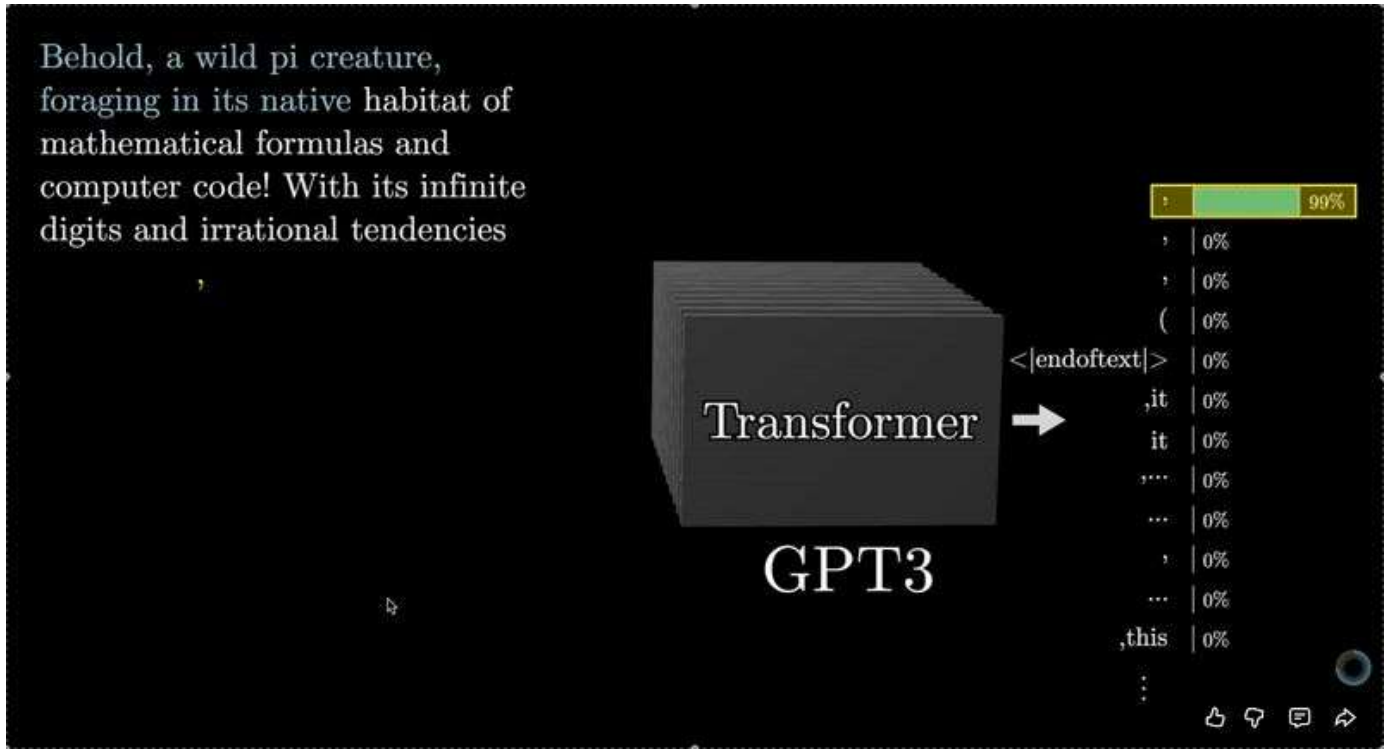
Fine-tuning is fast and efficient

- Adapt a pre-trained model to your specific task
- Requires much less data (hundreds to thousands of examples)
- Takes minutes to hours

GPT MODELS


Targets
x_2
x_3
x_4
...
Transformer
Inputs
x_1
x_2
x_3
...
Position ids
1
2
3
...
Next-token prediction

TEXT GENERATION: PREDICT, SAMPLE, REPEAT



COMPLETE THE REMAINING KAGGLE TASKS!